

## 14 Qualifications and Competence of Licensed Technical Persons

### 14.1 Essential Requirements

Every building/ development work for which permission is sought under the Code shall be planned, designed, and supervised by licensed persons. The licensed persons for carrying out the various activities shall be a) architect, (b) engineer, (c) structural engineer, (d) supervisor, (e) town planner, (f) landscape architect, (g) urban designer, and (h) utility service engineer. Requirements of registration for various professionals by the Authority or by the body governing such profession and constituted under a statute, as applicable to practice within the local body's jurisdiction, are given in 14.2.1 to 14.2.5. The competence of such licensed personnel to carry out various activities is also indicated in 14.2.1.2 to 14.2.5.2.

### 14.2 Qualifications and Competence of Licensed Persons

#### 14.2.1 Architect:

**14.2.1.1 Qualifications:** The minimum qualifications for an architect shall be the qualifications as provided for in the Architects Act, 1972 for registration with the Council of Architecture.

**14.2.1.2 Competence:** The licensed architect shall be competent to carry out the work related to the building/development permit as given below:

- a) All plans and information connected with building permit except engineering services of multistoried/special buildings.
- b) Issuing certificate of supervision and completion of all buildings pertaining to architectural aspects.
- c) Preparation of sub-division/layout plans and related information connected with development permit of area up to 1 hectare for metro-cities and 2 hectares for other places.
- d) Issuing certificate of supervision for development of land of area up to 1 hectare for metro-cities and 2 hectares for other places.

#### 14.2.2 Engineer

**14.2.2.1 Qualifications:** The minimum qualifications for an engineer shall be graduate in civil engineering/architectural engineering of recognized Indian or foreign university, or the Member of Civil Engineering Division/Architectural Engineering Division of the Institution of Engineers (India) or the statutory body governing such profession, as and when established.

**14.2.2.2 Competence:** The licensed engineer shall be competent to carry out the work related to the building/development permit as given below:

- a) All plans and information connected with building permit.
- b) Structural details and calculations of buildings on plot up to 500 sq.m and up to 5 storeys or 16.0 m in height.
- c) Issuing certificate of supervision and completion for all buildings.
- d) Preparation of all service plans and related information connected with development permit; and
- e) Issuing certificate of supervision for development of land for all area.

#### 14.2.3 Structural Engineer

**14.2.3.1 Qualifications:** The minimum qualifications for a structural engineer shall be graduate in civil engineering of recognized Indian or foreign university, or Corporate Member of Civil Engineering

Division of Institution of Engineers (India), and with minimum 3 years' experience in structural engineering practice with designing and field work.

NOTE- The 3 years' experience shall be relaxed to 2 years in the case of post-graduate degree of recognized Indian or foreign university in the branch of structural engineering. In case of doctorate in structural engineering, the experience required would be one year.

**14.2.3.2 Competence:** The licensed structural engineer shall be competent to prepare the structural design, calculations and details for all buildings and supervision.

A-2.3.1.1 In case of buildings having special structural features, as decided by the Authority, which are within the horizontal areas and vertical limits specified in A-2.2.1(b) and A-2.4.1(a) shall be designed only by structural engineers.

## 14.2.4 Supervisor

### 14.2.4.1 Qualifications

The minimum qualifications for a supervisor shall be diploma in civil engineering or architectural assistantship, or the qualification in architecture or engineering equivalent to the minimum qualification prescribed for recruitment to non-gazetted service by the Government of India plus 5 years' experience in building design, construction and supervision.

### 14.2.4.2 Competence

The licensed supervisor shall be competent to carry out the work related to the building permit as given below:

- a) All plans and related information connected with building permit for residential buildings on plot up to 100 m<sup>2</sup> and up to two storeys or 7.5 m in height; and
- b) Issuing certificate of supervision for buildings as per (a)

## 14.2.5 Town Planner

### 14.2.5.1 Qualifications

The minimum qualification for a town planner shall be graduate or post-graduate degree in town and country planning with Associate Membership of the Institute of Town Planners.

### 14.2.5.2 Competence

The licensed town planner shall be competent to carry out the work related to the development permit as given below.

- a) Preparation of plans for land sub-division/layout and related information connected with development permit for all areas.
- b) Issuing of certificate of supervision for development of land of all areas.

NOTE- However, for land layouts for development permit above 5 hectares in area, landscape architect shall also be associated, and for land development infrastructural services for roads, water supplies, sewerage/drainage, electrification, etc. the licensed engineers for utility services shall be associated.

## 14.2.6 Landscape Architect

### 14.2.6.1 Qualifications

The minimum qualification for a landscape architect shall be the bachelor's or master's degree in landscape architecture or equivalent from recognized Indian or foreign university.

### 14.2.6.2 Competence

The licensed landscape Architect shall be competent to carry out the work related to landscape design for building development permit for land areas 5 hectares and above. In case of metro cities, this limit of land area shall be 2 hectares and above.

**Note:** For smaller areas below the limits indicated above, association of landscape architect may also be considered from the point of view of desired landscape development.

### 14.2.7 Urban Designer

#### 14.2.7.1 Competence

The minimum qualification for an urban designer shall be master's degree in urban design or equivalent from recognized Indian or foreign university.

#### 14.2.7.2 Competence

The licensed urban designer shall be competent to carry out the work related to the building permit for urban design for land areas more than 5 hectares and campus area more than 2 hectares. He/She shall also be competent to carry out the work of urban renewal for all areas.

**Note:** For smaller areas below the limits indicated above, association of landscape architect may also be considered from the point of view of desired urban design.

### 14.2.8 Engineers for Utility Services

For multistoried/special buildings the work of building and plumbing services shall be executed under the planning, design and supervision of competent personnel. The qualification for licensed mechanical engineer (including HVAC), electrical engineer and plumbing engineers for carrying out the work of Air-conditioning, Heating and Mechanical Ventilation, Electrical Installations, Lifts and Escalators and Water Supply, Drainage. Sanitation and Gas Supply installation respectively shall be as given in Part 8 'Building Services' and Part 9 'Plumbing Services' of National Building Code, 2016 or as decided by the Authority taking into account practices of the National Professional bodies dealing with the specialist engineering services.

## 14.3 Builder/ Developer Entity

The minimum qualification and competence for the builder/developer entity for various categories of building and infrastructural development shall be as decided by the Authority to ensure compliance of quality, safety and construction practices as required under the National Building Code, 2016.

## 14.4 Qualifications of structural engineers in earthquake zone

**Chartered Structural Engineer:** Minimum experience (in years) to supervise the design and construction of multi-storey buildings and related infrastructure facilities.

| storey buildings and related infrastructure facilities.                                    |                                                                                                                                    |        |        |                                               |        |        |
|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------|--------|-----------------------------------------------|--------|--------|
| For multi-storey buildings/ critical infrastructure facilities                             | Building location in Earthquake Zone                                                                                               |        |        |                                               |        |        |
|                                                                                            | Zone-1                                                                                                                             | Zone-2 | Zone-3 | Zone-4                                        | Zone-5 | Zone-6 |
| up to a maximum 4-storeys or 12-meter height or 2500 sqm. floor area.                      | Graduate SE: 3 years<br>P.Graduate SE: 1 year                                                                                      |        |        | Graduate SE: 5 years<br>P.Graduate SE: 3 year |        |        |
| up to a maximum 8-storeys or 24-meter height or 5000 sqm. covered area.                    | Graduate SE: 7 years<br>P.Graduate SE: 5 year                                                                                      |        |        | Graduate SE: 9 years<br>P.Graduate SE: 7 year |        |        |
| with more than 8-storeys or height more than 24-meters or covered area more than 5000 sqm. | Graduate SE: 10 years; Post-graduate SE: 8 years.<br><br>The structural design shall be countersigned by a Professor of Structural |        |        |                                               |        |        |

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|  | Engineering of IIT Roorkee University or any other specified technical institute. |
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**Authorized Site Civil Engineer:** Minimum experience (in years) to supervise the design and construction of multi-storey buildings and related infrastructure facilities

| For multi-storey buildings/ critical infrastructure facilities                                | Building location in Earthquake Zone                        |        |        |                                                              |        |        |
|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------|--------|--------|--------------------------------------------------------------|--------|--------|
|                                                                                               | Zone-1                                                      | Zone-2 | Zone-3 | Zone-4                                                       | Zone-5 | Zone-6 |
| up to a maximum 4 storeys or 12 meters height or 2500 sqm. floor area**                       | Diploma Civil Engr: 6 years<br>Graduate Civil Engr: 3 years |        |        | Diploma Civil Engr: 10 years<br>Graduate Civil Engr: 5 years |        |        |
| up to a maximum 8 storeys or 24-meter height or 5000 sqm. covered area**                      | Diploma Civil Engr: NA*<br>Graduate Civil Engr: 6 years     |        |        | Diploma Civil Engr: NA*<br>Graduate Civil Engr: 10 years     |        |        |
| with more than 8 storeys or height more than 24 meters or covered area more than 5000 sqm. ** | Diploma Civil Engr: NA*<br>Graduate Civil Engr: 15 years    |        |        |                                                              |        |        |

\*NA – Not Authorized

\*\*Note: One site engineer of specified qualification shall be deployed to supervise every 2500 sqm.

**Inspecting Civil Engineer:** Minimum experience (in years) to supervise the design and construction of multi-storey buildings and related infrastructure facilities

| For multi-storey buildings/ critical infrastructure facilities                             | Building location in Earthquake Zone                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |        |                               |        |        |
|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|-------------------------------|--------|--------|
|                                                                                            | Zone-1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Zone-2 | Zone-3 | Zone-4                        | Zone-5 | Zone-6 |
| up to a maximum 4 storeys or 12 meters height or 2500 sqm. floor area.                     | Graduate Civil Engr: 5 years                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |        | Graduate Civil Engr: 7 years  |        |        |
| up to a maximum 8 storeys or 24-meter height or 5000 sqm. covered area.                    | Graduate Civil Engr: 8 years                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |        | Graduate Civil Engr: 10 years |        |        |
| with more than 8 storeys or height more than 24 meters or covered area more than 5000 sqm. | Graduate Civil Engr: 15 years.<br><br>A joint panel of two serving/retired experts shall be deployed to inspect these construction works, in which one expert should at least be of the level of Chief Engineer/equivalent in any engineering department/undertaking of the State/Central Government, and the other expert, as far as possible, should be in a specified technical field. The experts shall be of the level of Professor of Civil Engineering Department of the Institute, in whose absence both the experts shall be of the level of Chief Engineer/equivalent. |        |        |                               |        |        |

## 14.5 Licensed qualified third-party agency for fire and life safety systems

The minimum qualification and competence for the entities to act as ‘licensed qualified third-party agencies for installation and maintenance of fire-prevention and life safety systems’ shall be as decided by the Director General, Uttar Pradesh Fire and Emergency Services.

Licenses to act as qualified third-party agencies for fire-prevention and life safety systems shall be issued for Fire Department as per the provisions of Section-26 of the Uttar Pradesh Fire and Emergency Services Act, 2022 and corresponding rules.